

International Subarachnoid Aneurysm Trial of Neurosurgical Clipping Versus Endovascular Coiling

Subgroup Analysis of 278 Elderly Patients

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Background and Purpose—It is often thought that elderly patients in particular would benefit from endovascular aneurysm treatment. The aim of this analysis was therefore to compare the efficacy and safety of endovascular coiling (EVT) with neurosurgical clipping (NST) in the subgroup of elderly SAH patients in the International Subarachnoid Aneurysm Trial (ISAT).

Methods—In the ISAT cohort 278 SAH patients, 65 years or older, were enrolled. The patients were randomly allocated EVT (n=138) or NST (n=140). The primary outcome was the proportion of patients with a modified Rankin scale score of 0 to 2 (independent survival) at 1 year after the SAH. The rates of procedural complications and adverse events were also recorded.

Results—83 of 138 (60.1%) patients allocated EVT were independent compared to 78 of 140 (56.1%) allocated NST (N.S.). 36 of 50 (72.0%) patients with internal carotid and posterior communicating artery aneurysms allocated EVT were independent compared to 26 of 50 (52.0%) allocated NST ($P<0.05$). 10 of 22 (45.5%) patients with middle cerebral artery aneurysms allocated EVT were independent compared to 13 of 15 (86.7%) allocated NST ($P<0.05$). The epilepsy frequency was 0.7% in the EVT group compared to 12.9% in the NST group ($P<0.001$).

Conclusions—In good grade elderly SAH patients with small anterior circulation aneurysms, EVT should probably be the favored treatment for ruptured internal carotid and posterior communicating artery aneurysms, whereas elderly patients with ruptured middle cerebral artery aneurysms appear to benefit from NST. EVT resulted in a lower epilepsy frequency than NST. (*Stroke*. 2008;39:2720-2726.)

Key Words: subarachnoid hemorrhage ■ intracranial aneurysm ■ aged ■ endovascular treatment ■ neurosurgery ■ clinical trial

The management of elderly subarachnoid hemorrhage (SAH) patients is a clinical challenge. In conjunction with the improved health and the longer life expectancies now observed in the Western world, the proportion of elderly in the general population is substantial and increasing. The incidence of SAH increases with advancing age, reaching 78 per 100 000 per year in the eighth decade compared to 15 per 100 000 per year in the fourth to sixth decade of life.¹ Hence, the elderly account for a significant number of the patients presenting with SAH. Thirty years ago active management and neurosurgical treatment was not considered feasible in elderly patients, and conservative treatment was advocated in patients over 60 years.² The view that some elderly patients would benefit from neurosurgical treatment of the ruptured aneurysm³⁻⁵ was gradually adopted, and today treatment is considered even in patients more than 80 years old provided their prior quality of life was good.^{6,7} Nevertheless, elderly SAH patients have a greater risk of poor outcomes^{8,9} than

younger patients do. Reasons for this are less active management and conservative referral patterns,^{10,11} poorer clinical grades on admission,¹²⁻¹⁴ and a higher frequency of comorbidity,^{8,15} and that patients over 50 years of age tolerate craniotomy and clipping of intracranial aneurysms less well. Changes in referral patterns and more active management have improved outcome over time in elderly SAH patients.¹⁶

In 1990 the detachable platinum coil^{17,18} was introduced for obliterating ruptured aneurysms. Since 1995 endovascular treatment (EVT) has been widely used in patients with ruptured and unruptured aneurysms.^{19,20} With the prospect of reducing the risk of rebleeding without the need for craniotomy, thus reducing surgical trauma, EVT was conceived as a promising alternative to neurosurgical treatment (NST), especially in elderly and poor-grade patients. Some studies focusing on EVT of ruptured aneurysms specifically in elderly patients have shown favorable results.²¹⁻²⁶ However, the benefit of EVT versus conventional NST specifically in

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elderly patients with ruptured aneurysms has not been proven in a large randomized study.

The International Subarachnoid Aneurysm Trial (ISAT) is a randomized study undertaken to compare the safety and efficacy of EVT with NST in SAH patients suitable for both treatments. The first report from ISAT²⁷ showed that EVT reduced the relative risk of dependency or death at 1 year after aneurysm rupture with 22.6% compared to NST in patients of all ages. In a second article from ISAT²⁸ this finding was found consistent in all prespecified subgroups. Furthermore, although medium term rebleeding rates were low in both groups, rebleeding was slightly more common in the EVT group.

The present article reports on a subgroup analysis of patients 65 years or older in the ISAT cohort. The specific aims were to compare EVT with NST in regard to 1-year functional outcome, as well as procedural complication rates and frequency of adverse events in this age group.

Materials and Methods

Patients

The ISAT study protocol including randomization procedures and minimization criteria has previously been published.^{27,29} Eligible patients had SAH attributable to a ruptured aneurysm suitable for both EVT and NST. For this subgroup analysis, patients 65 years or older at the time of the hemorrhage were selected. Patients gave appropriate informed consent, according to the criteria laid down by the local ethics committee.

Study Procedures

Key baseline characteristics were recorded before the treatment allocation was issued. These were: age, sex, clinical grade on the World Federation of Neurological Surgeons (WFNS) scale,³⁰ size and location of the target aneurysm, and the extent of blood on the CT scan according to Fisher scale.³¹

Data were collected and reported on specific case record forms, regarding technical success and procedural complications for endovascular and neurosurgical procedures as well as any adverse events and additional procedures after the time of randomization during the first admission.

Procedural complications recorded during EVT were aneurysm rupture, parent artery occlusion, the use of thrombolytic agent and coil migration. During NST, the following procedural complications were recorded: aneurysm rupture, parent artery occlusion and the use of temporary clips. On waking the patient after anesthesia from either treatment, neurological deterioration was defined as the patient's WFNS grade being lower after than before the procedure or if a neurological deficit was present. If the deficit was thought to be of thrombo-embolic nature this was specifically recorded.

The recorded adverse events were pre- and postprocedural rebleeding, delayed ischemic deficit, hydrocephalus, epilepsy, infection, cerebral hematoma, pulmonary complications, serious groin hematoma, and severe headache.

Outcome

The main outcome measure was a modified Rankin scale³² assessed at 1 year after the hemorrhage. The modification of the original scale by Rankin and the full text of the questionnaire used to assess functional outcome are previously published in the trial protocol.²⁹

Data Management and Statistical Analyses

The data for this report were extracted from the ISAT database on November 9, 2005, and include some selected data on all 278 randomized patients aged 65 or older.

Comparisons between EVT and NST were made on an intention to treat basis. If a patient crossed over from one treatment modality to the other, they were analyzed in the group in which they were allocated based on the randomization result. However, comparison of procedural complications was made between the actual treatment groups.

In the outcome analyses modified Rankin scores were dichotomized into Rankin 0 to 2 (Independent survival) and Rankin 3 to 6 (Dependency or death). Categorical data were examined by χ^2 statistics. The Mann-Whitney *U* test was used to compare the nonnormally distributed data, and Student *t* test was used to compare normally distributed data. All analyses were done with commercially available software (StatSoft, Inc, 2005; STATISTICA data analysis software system, version 7.1; www.statsoft.com).

Role of the Funding Source

The sponsors of the study had no role in study design, data collection, data analysis, data interpretation, or writing the report.

Results

Of the 2143 patients entered in the ISAT cohort, 278 (13%) were 65 years or older at the time of the SAH and eligible for this subgroup analysis. 138 patients were randomized to EVT and 140 patients to NST. Baseline characteristics were available for all 278 patients. Sex, age, WFNS grade, Fisher grade, number of detected aneurysms, and maximum aneurysm lumen diameter are shown in Table 1.

There was no significant imbalance regarding the target aneurysm location between the two groups (Table 2). There was a tendency toward more left-sided anterior cerebral artery aneurysms in the NST group. In both the EVT and the NST groups, there were more right- than left-sided posterior communicating artery and middle cerebral artery aneurysms, but no difference between the two treatment groups.

Two hundred sixty-three (94.6%) patients received the treatment they were allocated. Five patients crossed over from NST to EVT before the first procedure, whereas none crossed over from EVT to NST. Two patients in the EVT group and 8 patients in the NST group were not treated for their ruptured aneurysm. Thus, 141 patients underwent EVT and 127 NST. One hundred twenty-nine (91.5%) of the EVT procedures were completed as intended compared to 119 (94.4%) of the neurosurgical procedures (N.S.). The reasons for crossovers, no treatment, and not completing the procedures are outlined in Table 3.

There was no difference between the two treatment groups in regard to time interval from SAH to randomization, from randomization to treatment procedure, or in the length of stay in intensive care unit. Patients allocated EVT stayed for a significantly shorter time in hospital (Mean: 19.4 days, Median: 15 days, Range: 1 to 84 days), than the patients allocated NST (Mean: 23.3 days, Median: 18 days, Range: 3 to 181 days; $P < 0.05$).

Outcome

Data on functional outcome at 1 year after the hemorrhage was available for 277 patients (99.6%). For one patient allocated NST, the outcome data were unavailable. Eighty-three (60.1%) of the 138 patients allocated EVT were independent at 1 year, compared with 78 (56.1%) of the 139 patients allocated NST (N.S.). The 1-year mortality

Table 1. Baseline Characteristics

	Endovascular (n=138)		Neurosurgery (n=140)		Total (n=278)		
	n	(%)	n	(%)	n	(%)	
Male sex	43	(31.2)	36	(25.7)	79	(28.4)	N.S.
Age, y							
65–69	77	(55.8)	75	(53.6)	152	(54.7)	N.S.
70–74	47	(34.1)	50	(35.7)	97	(34.9)	
75–79	12	(8.7)	13	(9.2)	25	(9.0)	
80–86	2	(1.4)	2	(1.4)	4	(1.4)	
WFNS grade							
1	73	(52.9)	82	(58.6)	155	(55.7)	N.S.
2	40	(29.0)	37	(26.4)	77	(27.7)	
3	11	(8.0)	11	(7.9)	22	(7.9)	
4	8	(5.8)	6	(4.3)	14	(5.0)	
5	3	(2.2)	3	(2.1)	6	(2.2)	
6 (not assessable)*	3	(2.2)	1	(0.7)	4	(1.4)	
Fisher grade							
1	5	(3.6)	12	(8.6)	17	(6.1)	N.S.
2	14	(10.1)	17	(12.2)	31	(11.2)	
3	56	(40.6)	60	(42.9)	116	(41.7)	
4	63	(45.7)	51	(36.4)	114	(41.0)	
No. of aneurysms detected							
1	112	(81.2)	118	(84.3)	230	(82.7)	N.S.
2	18	(13.0)	18	(12.9)	36	(12.9)	
3	4	(2.9)	4	(2.9)	8	(2.9)	
>4	4	(2.9)	0	(0)	4	(1.4)	
Maximum target aneurysm lumen (mm)							
<5	76	(55.1)	76	(54.3)	152	(54.6)	N.S.
6–10	49	(35.5)	54	(38.6)	103	(37.0)	
>11	13	(9.4)	10	(7.1)	23	(8.3)	

WFNS indicates World Federation of Neurological Surgeons clinical grading scale.

*Patient ventilated and clinical state could not be assessed.

rates were 18.1% for the EVT group and 21.6% for the NST group (N.S.).

There was no difference in outcome between EVT and NST for patients with ruptured aneurysms of the ACA and

Table 2. Aneurysm Locations by Randomized Treatment

	Endovascular		Neurosurgery		Total		
	n	(%)	n	(%)	n	(%)	
ACA-ACoM	65	(47.1)	71	(50.7)	136	(48.9)	N.S.
ICA	32	(23.2)	34	(24.3)	66	(23.7)	N.S.
PCoM	18	(13.0)	17	(12.1)	35	(12.6)	N.S.
MCA	22	(15.9)	15	(10.7)	37	(13.3)	
BA bifurcation	1	(0.7)	0	(0.0)	1	(0.4)	N.S.
PICA	0	(0.0)	3	(2.1)	3	(1.1)	N.S.
Subtotal	1	(0.7)	3	(2.1)	4	(1.4)	
Total	138		140		278		

ACA indicates anterior cerebral artery; ACoM, anterior communicating artery; ICA, internal carotid artery; PCoM, posterior communicating artery; MCA, middle cerebral artery; BA, basilar artery; PICA, posterior inferior cerebellar artery.

ACoM. The patients that received EVT for an ICA or PCoM aneurysm had better outcomes than patients who were neurosurgically treated. Conversely, the patients with MCA aneurysms who received NST had better outcomes than the EVT patients (Table 4). There was no difference in outcome between patients with left-sided or right-sided middle cerebral artery aneurysms in either treatment group.

Complications

Aneurysm rupture occurred more frequently during neurosurgical procedure than during EVT. There was no difference in the rate of neurological deterioration, thromboembolic problems, or parent artery occlusion between the groups (Table 5). Intraprocedural aneurysm rupture did not have any impact on the functional outcome in either treatment. In the EVT group, parent artery occlusion, thromboembolic problems, neurological deterioration, and the use of thrombolytic agents were associated with a greater risk of a dependent outcome. Procedural complications during neurosurgical treatment did not have a marked influence on the functional outcome.

Table 3. Technical Outcome of First Procedure

	Treatment Allocation		Total	
	Endovascular n	Neurosurgery n	n	(%)
Endovascular procedure				
Completed	124	5†	129	(91.5)
Failed to catheterize target aneurysm	3	0	3	(2.1)
Aneurysm catheterized but anatomy unsuitable	6	0	6	(4.3)
Not attempted*	3¶	0	3	(2.1)
Subtotal			141	
Neurosurgical procedure				
Clipped	0	119	119	(94.4)
Wrapped	0	3	3	(2.4)
Not completed (partial clipping or wrapping)	0	3	3	(2.4)
Not attempted*	0	2	2	(0.8)
Subtotal			127	
No procedure				
Died before procedure	2‡	4§	6	
Conservative treatment	0	4	4	
Total	138	140	278	

*Anesthesia started but procedure not completed—eg, because of change in patient's condition or presence of severe vasospasm preventing immediate EVT.

†Crossed over because of re-bleeding (1), poor grade (1), patient's preference (1), and unknown (2).

‡Re-bleeding (1) and clinical deterioration (1).

§Re-bleeding (4).

||Myocardial infarction (1), patient refused surgery (1), unknown (2).

¶Vasospasm (2), thromboembolic problem, and parent artery occlusion (1).

Adverse Events

There was a lower rate of epilepsy, infectious complications, and pulmonary complications in the patients allocated EVT than in the patients allocated NST. There were no differences between the two treatment groups in the frequency of meningitis, delayed neurological deficits, hydrocephalus, cerebral hematomas, groin hematomas, or severe headache during the first admission (Table 6).

Discussion

In this report, a subgroup of the 278 elderly patients within the ISAT cohort were analyzed to identify whether functional outcome and the rate of procedural complications and adverse events differed between the two treatment policies. The definition of elderly patients in this report was patients 65 years or older at the time of aneurysm rupture. This cut-off age defining the elderly is widely accepted in the Western world and has been used in several previous studies.^{14,16,23–25} This subgroup of the ISAT cohort was not prespecified and the study may therefore, statistically, not be considered a valid randomized study. However, we believe that much important and useful information can be drawn from the large amount of data gathered regarding this patient subgroup.

More than 90% of the patients were GCS $\geq$ 13 on admission, which is higher than in most published series. This

probably reflects the view in many centers that early neurosurgical aneurysm treatment was inappropriate in poor grade³³ elderly patients, and EVT was therefore preferred in these patients or unconscious elderly patients were not actively treated. This view may also account for the number of elderly patients entered the ISAT being lower than expected. The proportion of posterior circulation aneurysms was just over 1%, a low figure compared to other surgical and particularly endovascular series. Because of the high surgical risks and the good technical and functional outcome after EVT of posterior circulation aneurysms,³⁴ most centers probably considered EVT the favored modality for these surgically challenging aneurysms. Patients with middle cerebral artery aneurysms are also underrepresented in the study, which is probably because of often encountered unfavorable aneurysm anatomy for EVT. Neurosurgical therapy was likely considered the treatment of choice for these aneurysms in many centers.³⁵

The findings of the present study can be generalized to elderly patients similar to those enrolled, ie, good grade patients with a ruptured anterior circulation aneurysm with a suitable anatomy for either EVT or NST.

We were unable to find a significant difference in independent survival between EVT and NST, although there was a trend toward a greater proportion of independent survival after EVT (60.1%) than after NST (56.1%). The number of elderly patients in the ISAT study were relatively few, encompassing only 12.9% of the whole study population, thus not enough to reach the goal of the statistical power calculations set out in the ISAT study protocol. However, there is currently no reason to doubt that the reduction of dependent survival or death after EVT seen in all patients in the ISAT cohort should not be valid even in the elderly.

The location of the ruptured aneurysm had an important impact on functional outcome in the elderly patients in ISAT. Endovascular therapy of ICA and PComA aneurysms achieved a greater proportion of independent survival compared to the NST group. However, the patients who received NST for a ruptured MCA aneurysm had a greater chance of independent outcome compared to the EVT patients. These results suggest that EVT should be the favored treatment in elderly SAH patients with a ruptured aneurysm in the ICA and PComA territory, whereas neurosurgical clipping could be the treatment of choice for most MCA aneurysms in this patient subgroup. There was no indication that the difference in outcome between NST and EVT was the effect of imbalance between the treatment groups in terms of right- or left-sided aneurysms, WFNS grade, Fisher grade, patient age, aneurysm lumen diameter, or aneurysm neck size. However, the number of elderly patients with ruptured MCA aneurysms was relatively small, and the results must therefore be interpreted with caution. Careful selection of middle cerebral artery aneurysms suitable for EVT seems to be fundamental for good results.^{35–37}

A highly feared intraprocedural complication during both EVT and neurosurgical clipping of an intracranial aneurysm is aneurysm rupture. In large surgical series, aneurysm rupture during surgery is associated with less favorable outcome. In this series, the rate of aneurysmal rupture during

Table 4. Aneurysm Location and Outcome at 1 Year by Randomized Treatment

	Endovascular		Neurosurgery		Total		
Modified Rankin Scale	n	(%)	n	(%)	n	(%)	
ACA-ACoM aneurysm							
0–2 Independent survival	36	(55.4)	36	(50.7)	72	(52.9)	N.S.
3–6 Dependency or death	29	(44.6)	35	(49.3)	64	(47.1)	
	65		71		136		
ICA-PCoM aneurysm							
0–2 Independent survival	36	(72.0)	26	(52.0)	62	(62.0)	P<0.05
3–6 Dependency or death	14	(28.0)	24	(48.0)	38	(38.0)	
	50		50		100		
MCA aneurysm							
0–2 Independent survival	10	(45.5)	13	(86.7)	23	(62.2)	P<0.05
3–6 Dependency or death	12	(54.5)	2	(13.3)	14	(37.8)	
	22		15		37		
Subtotal	137		136		273		
All aneurysm locations							
0–2 Independent survival	83	(60.1)	78	(56.1)	161	(58.1)	N.S.
3–6 Dependency or death	55	(39.9)	61	(43.9)	116	(41.9)	
Total	138*		139†‡		277		

ACA indicates anterior cerebral artery; ACoM, anterior communicating artery; MCA, middle cerebral artery; ICA, internal carotid artery; PCoM, posterior communicating artery.

*138 randomized to endovascular, 1 basilar artery aneurysm.

†140 randomized to neurosurgery, 3 posterior inferior cerebellar artery aneurysms.

‡1 patient lost to follow-up: 1 PCoM aneurysm.

treatment was significantly higher in the neurosurgically treated group than in the EVT group. However, aneurysm rupture did not have a significant impact on outcome in either treatment group. The risk of thromboembolic problems was higher in the EVT group, and the occurrence of parent artery occlusion, thromboembolic problems, and use of thrombolytic agent had detrimental effects on outcome in this group. Further development of methods for avoiding inappropriate intraluminal thrombus formation during EVT is likely to improve outcome in these patients.

The frequency of epilepsy was greater after NST than after EVT in this cohort of elderly SAH patients. It is reasonable to believe that the craniotomy, aneurysm dissection, and the use of self-retaining brain retractors to some extent contribute to the development of epilepsy.

Table 5. Procedural Complications by Actual Treatment

	Endovascular (n=141)		Neurosurgery (n=127)		
	n	(%)	n	(%)	
Aneurysm rupture	9	(6.4)	21	(16.7)	$P<0.05$
Parent artery occlusion	7	(5.0)	4	(3.2)	N.S.
Thrombolytic agent used	7	(5.0)	...	...	
Coil migration	3	(2.1)	...	...	
Temporary clips used	...	...	37	(29.4)	
Neurological deterioration	16	(11.4)	24	(18.9)	N.S.
Thrombo-embolic problem	10	(7.1)	2	(1.6)	N.S.

Infections and pulmonary complications, particularly pneumonia, were found to be more prevalent in the patients allocated NST. These complications are well known in patients with prolonged artificial ventilation, increased length of stay in an intensive care unit, and prolonged bed rest. Accordingly we found that NST patients were admitted to hospital for a significantly longer period than the EVT patients, but there was no difference in the length of stay in intensive care unit. The difference in the frequency of infections and pulmonary complications between the two groups may be that EVT allowed quicker mobilization in these patients, and therefore prolonged bed rest could be avoided and earlier discharge from hospital was possible.

Concern has been raised that EVT does not allow irrigation of the basal cisterns and subarachnoid clot removal and therefore increase the risk of vasospasm and communicating hydrocephalus. In this study, there was no difference between the two treatment modalities in the frequency of delayed ischemic deterioration because of vasospasm or in the frequency of hydrocephalus necessitating a later shunting procedure. Thus, the importance of subarachnoid clot removal during craniotomy for aneurysm surgery may be overestimated in the elderly.

Summary

In this randomized cohort comprising primarily good grade elderly patients with small anterior circulation aneurysms considered suitable for both EVT and NST, a significant difference in independent survival at 1 year after the SAH

Table 6. Adverse Events by Randomized Treatment

	Endovascular (n=138)		Neurosurgery (n=140)		
	n	(%)	n	(%)	
Delayed neurological deficit	48	(34.8)	51	(36.4)	N.S.
Vasospasm	35	(25.4)	43	(30.7)	N.S.
Thrombo-embolic complication	13	(9.4)	8	(5.7)	N.S.
Hydrocephalus	35	(25.4)	38	(27.1)	N.S.
Epilepsy	1	(0.7)	18	(12.9)	$P<0.001$
Infection	16	(11.6)	29	(20.7)	$P<0.05$
Meningitis	2	(1.5)	3	(2.1)	N.S.
Other infection	14	(10.1)	26	(18.6)	$P<0.05$
Cerebral hematoma*	14	(10.2)	16	(11.4)	N.S.
ICH	6	(4.4)	11	(7.9)	N.S.
ECH	4	(2.9)	4	(2.9)	N.S.
Both ICH and ECH	4	(2.9)	1	(0.7)	N.S.
Pulmonary complications	26	(18.8)	44	(31.4)	$P<0.05$
Pneumonia	7	(5.1)	21	(15.0)	$P<0.01$
Pulmonary edema	8	(5.8)	11	(7.9)	N.S.
Other pulmonary complications	11	(8.0)	12	(8.6)	N.S.
Groin hematoma	1	(0.7)	0	(0.0)	N.S.
Severe headache	5	(3.6)	1	(0.7)	N.S.
Other adverse event	38	(27.7)	43	(30.7)	N.S.

*Missing data for 1 patient.

could not be demonstrated. However, a trend indicating better outcomes in the EVT group could be shown, which is in agreement with the results in the whole ISAT cohort with patients of all ages. Furthermore, elderly patients allocated EVT had a shorter length of stay in the hospital and had less infectious and pulmonary complications and a lower frequency of epilepsy than the patients allocated NST. The location of the ruptured aneurysm had a marked influence on the functional outcome depending on the treatment modality. EVT should be the treatment of choice for ruptured internal carotid artery and posterior communicating artery aneurysms in the elderly SAH patient. Elderly patients with ruptured middle cerebral artery aneurysms appear to benefit from neurosurgical clipping.

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